

Software | nand2tet... xNAND2Tetris xRAM512 Code Expl... x(4) WhatsApp xNand2tetris PC cod... x+Ask Gemini

nand2tetris.github.io/web-ide/chip/

NAND2Tetris / Hardware Simulator / PC.hdl

HDL

--LBS

Project 3

PC

Chip PC

Eval

Reset

Clock: 15

5 /**

9 * else if inc(t): out(t+1) = out(t) + 1

10 * else out(t+1) = out(t)

11 */

12 CHIP PC {

13 IN in[16], reset, load, inc;

14 OUT out[16];

15

16 PARTS:

17 Inc16(in=feedback, out=incremented);

18

19 Mux16(a=feedback, b=incremented, sel=inc, out=afterInc);

20 Mux16(a=afterInc, b=in, sel=load, out=afterLoad);

21 Mux16(a=afterLoad, b=false, sel=reset, out=afterReset);

22

23

24 Register(in=afterReset, load=true, out=out, out=feedback);

25 }

Input pins

in 0 1 0 1 0 1 1 0 1 1 0 0 1 1 0

reset 1

load 0

inc 0

Output pins

out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Internal pins

feedback 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

incremented 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1

Test

Steps: 32

Outputs: 31

Slow

Fast

Test Script

Compare File

Output File

Diff Table

115 output;

116

117 set in 22222,

118 set reset 1,

119 set inc 0,

120 tick,

121 output;

122

123 tock,

124 output;

Simulation successful: The output file is identical to the compare file

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NAND2Tetris / Hardware Simulator / RAM16K.hdl

HDL

--LBS

Project 3

RAM16K

Chip RAM16K

Eval

Reset

Clock: 81

```
10 */
11 CHIP RAM16K {
12   IN in[16], load, address[14];
13   OUT out[16];
14
15   PARTS:
16   DMux4Way(in=load, sel=address[12..13],
17     a=load0, b=load1, c=load2, d=load3);
18
19   RAM4K(in=in, load=load0, address=address[0..11], out=out0);
20   RAM4K(in=in, load=load1, address=address[0..11], out=out1);
21   RAM4K(in=in, load=load2, address=address[0..11], out=out2);
22   RAM4K(in=in, load=load3, address=address[0..11], out=out3);
23
24   Mux4Way16(
25     a=out0, b=out1, c=out2, d=out3,
26     sel=address[12..13], out=out);
27 }
```

Input pins

in 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 de

load 0

address 1 1 1 1 0 1 0 1 0 1 0 1 0 1 dec

Output pins

out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 de

Internal pins

load0 0

load1 0

load2 0

Test

Steps: 292

Outputs: 320

Slow

Fast

Test Script

Run

Compare File

Output File

Diff Table

```
1016 output;
1017 set address %0101101010101,
1018 eval,
1019 output;
1020 set address %0110101010101,
1021 eval,
1022 output;
1023 set address %0111101010101,
1024 eval,
1025 output;
```

Simulation successful: The output file is identical to the compare file

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NAND2Tetris / Hardware Simulator / RAM4K.hdl

HDL

--LBSt

Project 3

RAM4K

Chip RAM4K

Eval

Reset

Clock: 81

```
11 CHIP RAM4K {
16   DMux8Way(in=load, sel=address[9..11],
18     e=load4, f=load5, g=load6, h=load7);
19
20   RAM512(in=in, load=load0, address=address[0..8], out=out0);
21   RAM512(in=in, load=load1, address=address[0..8], out=out1);
22   RAM512(in=in, load=load2, address=address[0..8], out=out2);
23   RAM512(in=in, load=load3, address=address[0..8], out=out3);
24   RAM512(in=in, load=load4, address=address[0..8], out=out4);
25   RAM512(in=in, load=load5, address=address[0..8], out=out5);
26   RAM512(in=in, load=load6, address=address[0..8], out=out6);
27   RAM512(in=in, load=load7, address=address[0..8], out=out7);
28
29   Mux8Way16(
30     a=out0, b=out1, c=out2, d=out3,
31     e=out4, f=out5, g=out6, h=out7,
32     sel=address[9..11], out=out);
33 }
```

Input pins

in 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 de

load 0

address 1 1 1 1 0 1 0 1 0 1 0 1 dec

Output pins

out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 de

Internal pins

load0 0

load1 0

load2 0

Test

Steps: 292

Outputs: 320

Slow

Fast

Test Script

Compare File

Output File

Diff Table

```
1016 output;
1017 set address %01011010101,
1018 eval,
1019 output;
1020 set address %01101010101,
1021 eval,
1022 output;
1023 set address %01111010101,
1024 eval,
1025 output;
```

Simulation successful: The output file is identical to the compare file

Software | nand2tet... x

NAND2Tetris x

RAM512 Code Expl... x

(5) WhatsApp x

Nand2tetris PC cod... x

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nand2tetris.github.io/web-ide/chip/

NAND2Tetris / Hardware Simulator / Bit.hdl

HDL

--LBS

Project 3

Bit

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/3/a/Bit.hdl

5 /**

6 * 1-bit register:

7 * If load is asserted, the register's value is set to in;

8 * Otherwise, the register maintains its current value:

9 * if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)

10 */

11 CHIP Bit {

12 IN in, load;

13 OUT out;

14

15 PARTS:

16 Mux(a=outLoop, b=in, sel=load, out=muxOut);

17 DFF(in=muxOut, out=outLoop, out=out);

18 }

Chip Bit

Eval

Reset

Clock: 107

Input pins

in 1

load 0

Output pins

out 0

Internal pins

outLoop 0

muxOut 0

Test

Steps: 217

Outputs: 215

Slow

Fast

Test Script

Compare File

Output File

Diff Table

855 tock,

856 output;

857

858 set in 1,

859 set load 0,

860 tick,

861 output;

862

863 tock,

864 output;

Simulation successful: The output file is identical to the compare file

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nand2tetris.github.io/web-ide/chip/

NAND2Tetris / Hardware Simulator / Register.hdl

HDL

--LBSt

Project 3

Register

Chip Register

Eval

Reset

Clock: 74

11 CHIP Register {

12 IN in[16], load;

13 OUT out[16];

14

15 PARTS:

16 Bit(in=in[0], load=load, out=out[0]);

17 Bit(in=in[1], load=load, out=out[1]);

18 Bit(in=in[2], load=load, out=out[2]);

19 Bit(in=in[3], load=load, out=out[3]);

20 Bit(in=in[4], load=load, out=out[4]);

21 Bit(in=in[5], load=load, out=out[5]);

22 Bit(in=in[6], load=load, out=out[6]);

23 Bit(in=in[7], load=load, out=out[7]);

24 Bit(in=in[8], load=load, out=out[8]);

25 Bit(in=in[9], load=load, out=out[9]);

26 Bit(in=in[10], load=load, out=out[10]);

27 Bit(in=in[11], load=load, out=out[11]);

28 Bit(in=in[12], load=load, out=out[12]);

Input pins

in 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 dec

load 1

Output pins

out 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 dec

Test

Steps: 151

Outputs: 149

Slow

Fast

Test Script

Compare File

Output File

Diff Table

559

560 tick,

561 output;

562

563 set load 1,

564 tick,

565 output;

566

567 tick,

568 output;

Simulation successful: The output file is identical to the compare file

Software | nand2tet...

NAND2Tetris

RAM512 Code Expl...

(5) WhatsApp

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NAND2Tetris / Hardware Simulator / RAM8.hdl

HDL

--LBS

Project 3

RAM8

Chip RAM8

Eval

Reset

Clock: 46

5 /**

10 */

11 CHIP RAM8 {

12 IN in[16], load, address[3];

13 OUT out[16];

14

15 PARTS:

16 DMux8Way(in=load , sel=address , a=l0 , b=l1 , c=l2 , d=l3 , e=l4 , f=l5 , g=l6 , h=l7);

17 Register(in=in , load=l0 , out=r0);

18 Register(in=in , load=l1 , out=r1);

19 Register(in=in , load=l2 , out=r2);

20 Register(in=in , load=l3 , out=r3);

21 Register(in=in , load=l4 , out=r4);

22 Register(in=in , load=l5 , out=r5);

23 Register(in=in , load=l6 , out=r6);

24 Register(in=in , load=l7 , out=r7);

25 Mux8Way16(a=r0 , b=r1 , c=r2 , d=r3 , e=r4 , f=r5 , g=r6 , h=r7 , sel=address , out=out);

26 }

Input pins

in 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 de

load 0

address 1 1 1 dec

Output pins

out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 de

Internal pins

i0 0

i1 0

i2 0

Test

Steps: 160

Outputs: 173

Slow

Fast

Test Script

Compare File

Output File

Diff Table

550 set address 5,

551 eval,

552 output;

553 set address 6,

554 eval,

555 output;

556 set address 7,

557 eval,

558 output;

559

Simulation successful: The output file is identical to the compare file

Software | nan... xNAND2Tetris xHome | nand2 xRAM512 Code x(5) WhatsApp xNand2tetris PC x+Ask Gemini

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NAND2Tetris / Hardware Simulator / RAM512.hdl

HDL

--LBS

Project 3

RAM512

Chip RAM512

Eval

Reset

Clock: 81

```
5 /**
6  */
11 CHIP RAM512 {
12     IN in[16], load, address[9];
13     OUT out[16];
14
15     PARTS:
16     DMux8Way(in=load, sel=address[6..8],
17         a=load0, b=load1, c=load2, d=load3,
18         e=load4, f=load5, g=load6, h=load7);
19
20     RAM64(in=in, load=load0, address=address[0..5], out=out0);
21     RAM64(in=in, load=load1, address=address[0..5], out=out1);
22     RAM64(in=in, load=load2, address=address[0..5], out=out2);
23     RAM64(in=in, load=load3, address=address[0..5], out=out3);
24     RAM64(in=in, load=load4, address=address[0..5], out=out4);
25     RAM64(in=in, load=load5, address=address[0..5], out=out5);
26     RAM64(in=in, load=load6, address=address[0..5], out=out6);
27     RAM64(in=in, load=load7, address=address[0..5], out=out7);
28 }
```

Input pins

in 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 de

load 0

address 1 1 1 1 0 1 0 1 0 dec

Output pins

out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 de

Internal pins

load0 0

load1 0

load2 0

Test

Steps: 292

Outputs: 320

Slow

Fast

Test Script

Compare File

Output File

Diff Table

```
1017 set address %0101101010,
1018 eval,
1019 output;
1020 set address %0110101010,
1021 eval,
1022 output;
1023 set address %0111101010,
1024 eval,
1025 output;
1026
```

Simulation successful: The output file is identical to the compare file

Software | x

NAND2Tet | x

Home | na | x

RAM512 C | x

(5) Whats/ | x

Nand2tet | x

Convert JP | x

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Ask Gemini

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nand2tetris.github.io/web-ide/chip/

NAND2Tetris / Hardware Simulator / RAM64.hdl

HDL

--LBSt

Project 3

RAM64

Chip RAM64

Eval

Reset

Clock: 81

```
5 /**
11 CHIP RAM64 {
12   IN in[16], load, address[6];
13   OUT out[16];
14
15   PARTS:
16   DMux8Way(in=load , sel=address[3..5] , a=l0 , b=l1 , c=l2 , d=l3 , e=l4 , f=l5 , g=l6 , h=l7 );
17   RAM8(in=in , load=l0 , address=address[0..2] , out=r0 );
18   RAM8(in=in , load=l1 , address=address[0..2] , out=r1 );
19   RAM8(in=in , load=l2 , address=address[0..2] , out=r2 );
20   RAM8(in=in , load=l3 , address=address[0..2] , out=r3 );
21   RAM8(in=in , load=l4 , address=address[0..2] , out=r4 );
22   RAM8(in=in , load=l5 , address=address[0..2] , out=r5 );
23   RAM8(in=in , load=l6 , address=address[0..2] , out=r6 );
24   RAM8(in=in , load=l7 , address=address[0..2] , out=r7 );
25   Mux8Way16(a=r0 , b=r1 , c=r2 , d=r3 , e=r4 , f=r5 , g=r6 , h=r7 , sel=address[3..5] , out=out );
26 }
```

Input pins

in 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 de

load 0

address 1 1 1 1 0 1 dec

Output pins

out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 de

Internal pins

i0 0

i1 0

i2 0

Test

Steps: 292

Outputs: 320

Slow

Fast

Test Script

Compare File

Output File

Diff Table

```
1016 output;
1017 set address %B101101,
1018 eval,
1019 output;
1020 set address %B110101,
1021 eval,
1022 output;
1023 set address %B111101,
1024 eval,
1025 output;
```

Simulation successful: The output file is identical to the compare file